

Advanced Human-Computer Interaction

Ubiquitous Computing, AR/VR, and Emotional Interaction Trends

Detailed Study Notes Explanations

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1 Part 1: Ubiquitous Computing & Augmented Realities

1.1 1. Introduction to Advanced Interaction

Traditional Human-Computer Interaction (HCI) focused on a user sitting in front of a desktop computer. Modern HCI challenges this by moving computation into the environment.

- **Ubiquitous Computing (Ubicomp):** The concept of filling the real world with computers so they become part of the background.
- **Virtual Reality (VR) & Augmented Reality (AR):** The concept of recreating the real world inside a computer (VR) or overlaying the computer onto the real world (AR).

1.2 2. Ubiquitous Computing (Ubicomp)

1.2.1 Core Philosophy

The term was coined by **Mark Weiser** at Xerox PARC in the early 1990s. He challenged the assumption of what a computer looks like.

"The most profound technologies are those that disappear. They weave themselves into the fabric of everyday life until they are indistinguishable from it."
— Mark Weiser

Weiser argued that the 1990s personal computer (a box on a desk) was **not** the ultimate form of computing. Instead, he envisioned a world where computing is:

- **Implicit:** You don't "use" a computer; you perform a task, and the computer assists invisibly.
- **Perceptual:** The system senses the environment (Context Awareness).

1.2.2 The Scale of Devices (Weiser's Classification)

Weiser proposed three scales of devices that correspond to traditional physical objects:

1. The Inch (Tabs/Pads):

- *Concept:* Small, pocket-sized devices.
- *Modern Equivalents:* Smartphones (iPhone, Android), Smartwatches, PDAs, Voice Recorders.
- *Usage:* Individuals own many of them. They communicate with each other and the environment.

2. The Foot (Notebooks/Tablets):

- *Concept:* Slate-sized devices, similar to a sheet of paper or a notebook.

- *Modern Equivalents:* iPads, Galaxy Tabs, Microsoft Surface, Digital Paper.
- *Usage:* An individual owns several, but they are often shared or left in specific locations (e.g., a "kitchen tablet").

3. The Yard (Boards):

- *Concept:* Large displays, similar to a whiteboard or bulletin board.
- *Modern Equivalents:* Interactive Whiteboards (SmartBoards), Large Plasma displays in public spaces.
- *Usage:* Buildings or institutions own them. They are shared resources used for collaboration.

Real-World Example: IoT (Internet of Things)

Weiser's vision is realized today through IoT. A "Smart Home" contains:

- **Inch:** Smart thermostats (Nest), smart bulbs.
- **Foot:** A tablet controlling the home hub.
- **Yard:** A smart TV acting as a dashboard.

1.3 3. Interaction Paradigms in Ubicomp

1.3.1 Implicit Input & Context Awareness

Ubicomp moves away from explicit input (keyboard/mouse) toward **Implicit Input**.

- **Sensor-based:** Uses accelerometers, GPS, light sensors, and thermometers.
- **Awareness:** The system knows "where" it is and "who" is there.
- **Challenge:** Introduces **Ambiguity**. Recognizers (voice, gesture) are probabilistic and not perfect (e.g., Siri misunderstanding a command).

1.3.2 Ambient and Peripheral Output

Displays shouldn't always demand full attention.

- **Ambient Displays:** Use physical environment features to signal information subtly.
- **Peripheral:** Designed to sit in the background of human attention.

Real-World Example: The Dangling String (Natalie Jeremijenko)

An 8-foot plastic string hangs from the ceiling. A motor connected to the ethernet cable spins the string.

- **Heavy Traffic:** The string spins wildly (visual/audible hum).
- **No Traffic:** The string is still.
- **Result:** Workers know the network status peripherally without checking a monitor.

1.4 4. Case Studies in Ubicomp

1.4.1 Ambient Wood

A project exploring how digital technology can augment learning in a natural environment (a forest).

- **Setup:** Real forest filled with electronics.
- **Tools:** Light and moisture meters recording data with GPS locations.
- **Interaction:**
 - *The Periscope:* A device that shows "invisible" things (e.g., biological processes) when pointed at RFID-tagged trees.
 - *Triggered Sound:* Moving through specific zones triggers audio explanations.
- **Goal:** Merging the physical exploration of nature with digital information layers.

1.4.2 City: Shared Experience (Mackintosh Interpretation Centre)

- **Scenario:** A mixed reality museum experience.
- **Participants:** Some visitors are physically present; others visit via Web or VR.
- **Interaction:** They can talk via microphones and "see" virtual avatars of one another.
- **Key Concept:** Shared experience across different modalities and locations.

1.5 5. Virtual Reality (VR) vs. Augmented Reality (AR)

1.5.1 Virtual Reality (VR)

VR is the total replacement of the physical world with a computer-generated simulation.

- **Goal:** Immersion.
- **Tech:** Headsets (HMDs) with small LED screens for each eye (stereoscopic 3D).
- **Input:** DataGloves (measure hand position), Body suits, Eye-gaze tracking.
- **Types:**
 1. **Headset VR:** Expensive, high immersion (e.g., Oculus Quest).
 2. **Desktop VR:** Uses standard monitor, cheap, less immersive.
 3. **Cave/Command & Control:** Scenes projected on walls, realistic environment, often used in military/flight simulation.

1.5.2 Augmented Reality (AR)

AR is the **superimposition** of digital information onto the real world.

- **Tech:** Semi-transparent goggles (HoloLens), Smartphone cameras (Pokemon Go), Head-Up Displays (HUDs) in cars/aircraft.
- **Applications:**
 - **Maintenance:** Projecting repair instructions directly onto a broken photocopier engine (arrows pointing to specific screws).
 - **Medical:** Overlaying MRI scan data onto a patient's body during surgery.
- **Key Issue - Registration:** The critical technical challenge of aligning the virtual object perfectly with the real-world object (e.g., the arrow must point exactly to the screw, not 2 inches to the left).

1.6 6. Data Visualization in VR/3D

Visualizing complex data structures (hierarchies, networks) benefits from 3D space.

- **Cone Trees:** Developed by Xerox. Instead of a 2D org chart which runs out of horizontal space, a Cone Tree arranges hierarchies in a 3D rotating cone.
- **Perspective Wall:** Bends a 2D linear view (like a timeline) so the center is detailed, and the sides are compressed (focus + context).
- **Virtual Wind Tunnel:** Visualizing invisible airflows using virtual "bubbles" or smoke streams in a 3D simulation, allowing engineers to walk around the airflow.

2 Part 2: Emotional Interaction (2024-2025 Trends)

2.1 1. Definition

Emotional Interaction is the branch of HCI where systems sense, interpret, or respond to human emotions. It has evolved from "Affective Computing" (simply processing data) to including ethics, culture, and user agency.

Feelings vs. Moods vs. Emotions

- **Emotions:** Short duration (seconds), high intensity, specific cause (e.g., Anger at an insult).
- **Feelings:** Conscious mental reactions to emotions.
- **Moods:** Long duration (hours/days), lower intensity, cause often unclear.

2.2 2. Six Key Trends (2024-2025)

2.2.1 Trend 1: Culture-Aware Emotional Intelligence

Emotion is not universal. A smile might mean happiness in the US but embarrassment in parts of Asia.

- **Problem:** Previous AI used "one-size-fits-all" emotion models (usually Western-centric).
- **Solution: Affective-CARA Framework.** It uses knowledge graphs to interpret emotion based on cultural context.

2.2.2 Trend 2: Emotion Support vs. Regulation

- **Shift:** Moving away from systems that try to "fix" or "control" a user's emotion (Regulation).
- **Goal:** Systems that **support** emotional expression and validate feelings (Agency).
- **Example:** Instead of a watch saying "You are stressed, calm down," it might ask "Do you want to take a break?" allowing the user to decide.

2.2.3 Trend 3: Emotional Output Ethics (R-CAGE Model)

If an AI sounds **too** sad or **too** happy, it can manipulate the user.

- **R-CAGE:** A model to guide AI emotional expression (Tone, Pacing, Intensity).
- **Ethical Goal:** Prevent emotional manipulation in chatbots and virtual agents.

2.2.4 Trend 4: Emotion-Aware Companions

- **Example: Livia** (an AR-based emotionally adaptive companion).
- **Tech:** Uses multimodal detection (voice + face) and memory compression to "remember" past interactions, building a relationship.
- **Use Case:** Combating loneliness in elderly populations.

2.2.5 Trend 5: Immersive Empathic Computing

Using VR to force users to "walk in someone else's shoes."

- **Tech:** VR headsets dominating research.
- **Model:** Uses Dimensional Models (Valence/Arousal) rather than just labeling emotions "Happy/Sad."
- **Risk:** High emotional presence can be traumatic if not managed ethically.

2.2.6 Trend 6: Advances in Affective Sensing

Moving beyond just facial recognition.

- **Multimodal Integration:** Combining physiological signals (heart rate, skin conductance/GSR) with behavioral signals (typing speed, voice pitch).
- **Application:** Adaptive learning systems that detect when a student is frustrated and lower the difficulty automatically.

2.3 3. Design Implications

1. **User Agency:** Systems must respect the user's right to feel. Do not force happiness.
2. **Transparency:** Users must know *when* their emotions are being sensed and *how* that data is used.
3. **Cultural Sensitivity:** Designs must adapt emotional displays (color, avatars, language) to the user's culture.

2.4 4. Class Activity Suggestion

Task: Redesign a chatbot to express emotion ethically.

- **Constraint 1:** Must follow R-CAGE pacing (don't respond instantly with high emotion).
- **Constraint 2:** Adapt emotional tone for a specific culture (e.g., High Context vs. Low Context cultures).